Exercise Sheet

Lecture 1

Vectors Spaces and Subspaces

Unless stated otherwise, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$ and core problems are finite dimensional. Displayed function and sequence spaces are examples or bridges. Tags in parentheses indicate the flavour of a problem; a $\star$ marks a harder one.

Core tutorial route

Work **1.1, 1.6, 1.8, 1.10, 1.18, 1.19, 1.24, and 1.25** in that order (about 65–80 minutes before discussion). This eight-problem route checks calculation, an axiom diagnosis, polynomial/sequence/function/matrix examples, scalar-field dependence, and both computational and explanatory direct-sum work.

Additional practice: 1.2–1.5, 1.7, 1.9, 1.11, 1.13–1.17, 1.21–1.23, 1.27. **Bridge:** 1.12 (ODE solution spaces). **Extension:** 1.20 and 1.26.

Warm-up and axioms

Exercise 1.1 (computational) In $\mathbb{C}^2$, let $u = (1+i,\ 2)$ and $w = (3,\ -i)$. Compute $u + w$, the vector $(2-i)\,u$, and the additive inverse $-u$.

Answer. $u + w = (4 + i,\ 2 - i)$; $\quad (2 - i)u = (3 + i,\ 4 - 2i)$; $\quad -u = (-1 - i,\ -2)$.

Exercise 1.2 (computational) In $\mathbb{R}^3$, let $u = (2, -1, 3)$ and $w = (0, 4, -2)$. Compute $u + w$, $3u$, and $2u - w$.

Answer. $u + w = (2, 3, 1)$; $\quad 3u = (6, -3, 9)$; $\quad 2u - w = (4, -6, 8)$.

Exercise 1.3 (computational) In $\mathcal{P}_3(\mathbb{R})$, let $p = 2 - t + 3t^2$ and $q = 1 + 4t - t^2 + t^3$. Compute $p + q$, $3p$, and $2p - q$.

Answer. $p + q = 3 + 3t + 2t^2 + t^3$; $\quad 3p = 6 - 3t + 9t^2$; $\quad 2p - q = 3 - 6t + 7t^2 - t^3$.

Exercise 1.4 (computational) In $M_2(\mathbb{R})$, let $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & -1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 3 & -1 \\ 4 & 2 \end{smallmatrix}\right)$. Compute $A + B$, $2A$, and $A - 2B$.

Answer. $A + B = \left(\begin{smallmatrix} 4 & 1 \\ 4 & 1 \end{smallmatrix}\right)$; $\quad 2A = \left(\begin{smallmatrix} 2 & 4 \\ 0 & -2 \end{smallmatrix}\right)$; $\quad A - 2B = \left(\begin{smallmatrix} -5 & 4 \\ -8 & -5 \end{smallmatrix}\right)$.

Exercise 1.5 (proof) Using only the vector space axioms, prove that $a\,v = 0$ (with $a \in \mathbb{F}$, $v \in V$) implies $a = 0$ or $v = 0$.

Exercise 1.6 (proof) Decide, with proof, whether $\mathbb{R}^2$ with the *usual* addition but the scalar multiplication $a \odot (x, y) := (ax,\ 0)$ is a vector space over $\mathbb{R}$.

Exercise 1.7 (proof) Prove the cancellation law in a vector space: if $u, v, w \in V$ satisfy $u + w = v + w$, then $u = v$. Use only the axioms.

Recognising vector spaces

Exercise 1.8 (computational) Which of the following are vector spaces under the natural operations?

1. the polynomials in $\mathcal{P}(\mathbb{R})$ of degree *exactly* 3;
2. the polynomials $p \in \mathcal{P}_3(\mathbb{R})$ with $p(1) = 0$;
3. the real sequences (x_n) with $x_n \to 0$.

Answer. (1) No (not closed under +; also lacks 0). (2) Yes. (3) Yes.

Exercise 1.9 (computational) Which of the following subsets of $\mathbb{R}^2$ are vector spaces under the usual operations?

1. $\{(x, y) : 2x - 3y = 0\}$;
2. $\{(x, y) : x \geq 0\}$;
3. $\{(x, y) : xy = 0\}$.

Answer. (1) Yes. (2) No (not closed under scaling by -1). (3) No (not closed under +).

Exercise 1.10 (computational) Which of the following subsets of $\mathbb{R}^{\mathbb{R}}$ (the real functions of one real variable) are vector spaces?

1. $\{f : f(3) = 0\}$;
2. $\{f : f(3) = 1\}$;
3. $\{f : f \text{ is bounded}\}$.

Answer. (1) Yes. (2) No (zero function fails $f(3) = 1$). (3) Yes.

Exercise 1.11 (computational) Which of the following subsets of the space of real sequences (x_n) are vector spaces?

1. $\{(x_n) : x_1 = x_2\}$;
2. $\{(x_n) : \text{only finitely many } x_n \text{ are nonzero}\}$;
3. $\{(x_n) : x_n \to 5\}$.

Answer. (1) Yes. (2) Yes. (3) No (zero sequence converges to $0 \neq 5$).

Exercise 1.12 (proof) Inside the vector space of twice-differentiable functions $y\colon \mathbb{R} \to \mathbb{R}$, show that the solutions of $\ddot{y} + 5\dot{y} + 6y = 0$ form a vector space, but the solutions of $\ddot{y} + 5\dot{y} + 6y = t$ do not.

Hint. Differentiation is linear, so check closure for the homogeneous equation. For the inhomogeneous one, ask whether the zero function is a solution.

Exercise 1.13 (proof) Prove that $\{p \in \mathcal{P}(\mathbb{R}) : p(1) = p(2)\}$ is a subspace of $\mathcal{P}(\mathbb{R})$, but that $\{p \in \mathcal{P}(\mathbb{R}) : p(1)\,p(2) = 0\}$ is not.

Subspaces

Exercise 1.14 (computational) For each subset of $\mathbb{F}^3$, decide whether it is a subspace.

1. $\{(x_1, x_2, x_3) : x_1 + 2x_2 + 3x_3 = 0\}$;
2. $\{(x_1, x_2, x_3) : x_1 + 2x_2 + 3x_3 = 4\}$;
3. $\{(x_1, x_2, x_3) : x_1 x_2 x_3 = 0\}$.

Answer. (1) Subspace. (2) Not a subspace (0 fails the condition). (3) Not a subspace (not closed under addition).

Exercise 1.15 (computational) For each subset of $\mathbb{R}^3$, decide whether it is a subspace.

1. $\{(x, y, z) : 2x - y = 0 \text{ and } z = 0\}$;
2. $\{(x, y, z) : x^2 = y^2\}$;
3. $\{(t, 2t, 3t) : t \in \mathbb{R}\}$.

Answer. (1) Subspace. (2) Not a subspace. (3) Subspace.

Exercise 1.16 (computational) For each subset of $\mathbb{R}^4$, decide whether it is a subspace.

1. $\{(x_1, x_2, x_3, x_4) : x_1 - x_2 + x_3 - x_4 = 0\}$;
2. $\{(x_1, x_2, x_3, x_4) : x_1 = x_4\}$;
3. $\{(x_1, x_2, x_3, x_4) : x_1 + x_2 = 1\}$.

Answer. (1) Subspace. (2) Subspace. (3) Not a subspace (0 fails the condition).

Exercise 1.17 (computational) Decide whether each set is a subspace of the indicated space.

1. $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$ in $\mathcal{P}_3(\mathbb{R})$;
2. $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 1\}$ in $\mathcal{P}_3(\mathbb{R})$;
3. $\{A \in M_2(\mathbb{R}) : a_{11} = a_{22}\}$ in $M_2(\mathbb{R})$.

Answer. (1) Subspace. (2) Not a subspace (0 fails). (3) Subspace.

Exercise 1.18 (proof) Prove that the trace-zero matrices $\{A \in M_{n\times n}(\mathbb{F}) : \operatorname{tr} A = 0\}$ form a subspace of $M_{n\times n}(\mathbb{F})$, and that the invertible matrices do *not*.

Exercise 1.19 ($\star$) Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$? Is it a subspace of $\mathbb{C}^2$ regarded as a *real* vector space?

Hint. The deciding operation is multiplication by the complex scalar i: check whether $i \cdot (1, 0)$ stays in $\mathbb{R}^2$.

Exercise 1.20 (proof) Let U and W be subspaces of a vector space V. Prove that $U \cup W$ is a subspace of V if and only if $U \subseteq W$ or $W \subseteq U$.

Hint. One direction is immediate. For the other, argue by contradiction: if neither contains the other, pick $u \in U \setminus W$ and $w \in W \setminus U$ and examine $u + w$.

Sums and direct sums

Exercise 1.21 (computational) In $M_2(\mathbb{R})$, write $A = \left(\begin{smallmatrix} 4 & 1 \\ 7 & 2 \end{smallmatrix}\right)$ as the sum of a symmetric and an antisymmetric matrix.

Answer. $A = \left(\begin{smallmatrix} 4 & 4 \\ 4 & 2 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & -3 \\ 3 & 0 \end{smallmatrix}\right)$.

Exercise 1.22 (computational) In $M_2(\mathbb{R})$, write $A = \left(\begin{smallmatrix} 3 & 5 \\ 1 & 7 \end{smallmatrix}\right)$ as the sum of a symmetric and an antisymmetric matrix.

Answer. $A = \left(\begin{smallmatrix} 3 & 3 \\ 3 & 7 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & 2 \\ -2 & 0 \end{smallmatrix}\right)$.

Exercise 1.23 (computational) In $M_3(\mathbb{R})$, write $A = \left(\begin{smallmatrix} 2 & 4 & 0 \\ 0 & 2 & 6 \\ 4 & -2 & 2 \end{smallmatrix}\right)$ as the sum of a symmetric and an antisymmetric matrix.

Answer. $A = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} + \begin{pmatrix} 0 & 2 & -2 \\ -2 & 0 & 4 \\ 2 & -4 & 0 \end{pmatrix}$.

Exercise 1.24 (computational) In $\mathbb{R}^3$, for each pair of subspaces decide whether the sum $U + W$ is direct.

1. $U = \{(x, y, 0)\}$ and $W = \{(0, 0, z)\}$;
2. $U = \{(x, y, 0)\}$ and $W = \{(x, 0, z)\}$.

Answer. (1) Direct ($U \cap W = \{0\}$, and $U \oplus W = \mathbb{R}^3$). (2) Not direct ($U \cap W = \{(x, 0, 0)\} \neq \{0\}$).

Exercise 1.25 (proof) Let U_e and U_o be the even and odd functions in $\mathbb{R}^{\mathbb{R}}$ (so $f(-x) = f(x)$ and $g(-x) = -g(x)$, respectively). Show that $\mathbb{R}^{\mathbb{R}} = U_e \oplus U_o$.

Exercise 1.26 ($\star$) Give three subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ such that $V_i \cap V_j = \{0\}$ for every $i \neq j$, yet $V_1 + V_2 + V_3$ is *not* direct. (This shows pairwise-trivial intersections are not enough for three or more subspaces.)

Hint. Pairwise-trivial intersection is weaker than directness — try three distinct lines lying in a single plane through the origin.

Exercise 1.27 (proof) Let U and W be subspaces of a vector space V. Prove that $U + W = W$ if and only if $U \subseteq W$.